

Proposed Solution

Date	30 July 2026
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Project Name	Intergalactic Space Travel Request (ISTR) System
Maximum Marks	4 Marks

1. Problem Statement

Traditional mission request processes rely heavily on manual documentation, email communication, and multiple disconnected approval stages, resulting in a slow approval process, lack of transparency, poor tracking, repetitive paperwork, communication delays, difficult reporting, and security concerns.

2. Proposed Solution

ISTR is a centralized ServiceNow application, built on App Engine Studio, that provides:

- Online Mission Requests – submitted through the Service Catalog
- Automated Approval Workflow – orchestrated via Flow Designer
- Role-Based Access – enforced through ACLs, Roles, and Groups
- Email Notifications – on submission, approval, rejection, and completion
- Dashboard – real-time Mission Analytics via Performance Analytics
- Reports – Mission Status, Approval Summary, Risk Analysis, Monthly Requests
- Mission Tracking – full lifecycle visibility from Draft to Closed
- Secure Data Management – custom table extending Task, protected by 10 ACLs

3. Objectives

- Automate mission request submission
- Eliminate manual paperwork
- Streamline approval workflows across departments
- Improve transparency for all stakeholders
- Provide centralized mission management
- Reduce approval delays
- Increase data and process security
- Improve reporting and analytics capability

4. Target Users

Role	Description
Mission Requester	Scientist, Astronaut, or Research Officer submitting requests
Manager	Reviews and gives first-level approval on mission requests
Finance Officer	Approves mission budget

Role	Description
Mission Control	Provides final mission approval and scheduling
System Administrator	Maintains and configures the application

5. Core Modules

#	Module	Implemented Using
1	Mission Request Management (Create/Edit/Cancel/View)	Service Catalog, Custom Table
2	Approval Management (Manager, Finance, Mission Control)	Flow Designer
3	Workflow Automation	Flow Designer
4	Business Logic	Business Rules
5	Client Validation	Client Scripts
6	Dynamic Forms	UI Policies
7	Security	ACLs, Roles, Groups
8	Notifications	Email Engine
9	Dashboard	Performance Analytics
10	Reports	ServiceNow Reports

6. Expected Benefits

- Faster mission request turnaround through automated multi-level approvals
- Full transparency and traceability of every mission request
- Reduced administrative overhead and manual follow-ups
- Stronger data security through granular ACL-based access control
- Real-time visibility into mission status and departmental workload via dashboards

7. Conclusion

The proposed ISTR solution transforms a manual, fragmented mission request process into a secure, transparent, and fully automated ServiceNow application — improving speed, accountability, and visibility across every stage of the mission lifecycle.